

Paper : DSE-A-2
(Operation Research)
Full Marks : 50

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
 - (a) What is surplus variables?
 - (b) When is an L.P.P. said to be a degenerate problem?
 - (c) Give an example of a two-person zero-sum problem.
 - (d) Define the pay-off matrix of a general zero-sum game.
 - (e) What do you understand by the basic feasible solutions of an L.P.P.?
 - (f) Define a convex set in an Euclidean Space.
 - (g) What do you understand by the objective of an L.P.P.?
 - (h) What do you understand by assignment problem?

2. Determine all basic feasible solutions of the set of equations :

$$2x_1 + 6x_2 + 2x_3 + x_4 = 3$$

$$6x_1 + 4x_2 + 4x_3 + 6x_4 = 2$$

$$x_1, x_2, x_3, x_4 \geq 0.$$
 Identify the basic and non-basic variables in each case. 10

3. (a) Food X contains 6 units of vitamin A per gram and 7 units of vitamin B per gram and cost 12 paise per gram. Food Y contains 8 units of vitamin A per gram and 12 units of vitamin B per gram and cost 20 paise per gram. The daily minimum requirement of vitamin A and B are 100 units and 120 units respectively. In finding the minimum cost of product mix, formulate the problem as a linear programming problem.

 (b) Solve the above L.P.P. graphically. 6+4

4. Use Big-M method to solve the following L.P.P : 10

$$\text{Maximize } Z = 2x_1 + x_2 + 3x_3$$
 subject to $x_1 + x_2 + 2x_3 \leq 5$

$$2x_1 + 3x_2 + 4x_3 = 12$$

$$x_1, x_2, x_3 \geq 0.$$

5. (a) Define primal and dual solutions with examples.

 (b) Prove that for Linear Programming Problems the dual of a dual problem is the primal problem. (2+2)+6

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(0785+0786+0787)

6. Consider the following data for a transportation problem :

	D_1	D_2	D_3	D_4	
S_1	4	12	8	10	17
S_2	9	16	10	6	7
S_3	7	5	13	15	8
S_4	8	14	11	12	8
	10	9	11	10	

- (a) Indicate a starting basic solution by north-west corner rule.
 (b) Find the optimal solution. 4+6
7. (a) Define : Assignment Problem.
 (b) A marketing manager has 5 salesmen and 5 sales districts. Considering the capabilities of the salesmen and the nature of districts, the marketing manager estimates that sales per month (in hundred rupees) for each salesman in each district would be as follows :

	1	2	3	4	5
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

Find the assignment of salesmen to districts that will result in maximum sale.

2+8

8. (a) Explain how saddle points are obtained in a game.
 (b) Solve the game :

	B1	B2
A1	-6	7
A2	4	-5
A3	-1	-2
A4	-2	5
A5	7	-6

3+7